

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Participant Name: Sahil Pulikal
- b. Problem Statement: AI-powered intelligent eldercare assistant that helps family caregivers manage medications, hospital discharge instructions, and daily care routines using Google Gemini multimodal AI.

Brief about the idea

CareCompass AI is an intelligent eldercare management platform that assists family caregivers in coordinating medications, daily schedules, and patient care.

The solution uses **Google Gemini's multimodal capabilities** to understand prescription labels, hospital discharge documents, and caregiver conversations. Instead of manually reading medical documents and creating reminders, caregivers simply upload a photo or PDF and the AI extracts relevant information, updates medications, and builds an optimized daily care plan.

The platform reduces medication errors, improves adherence, and significantly lowers the cognitive burden on caregivers while improving the quality of life for elderly patients.

We approached the problem by building an AI-first caregiver platform rather than a traditional medication reminder application.

The solution leverages Google Gemini for:

- AI Caregiver Chat Assistant
- Prescription OCR
- Hospital discharge document understanding
- AI-powered schedule optimization
- Intelligent medical question answering

The frontend provides a modern dashboard while the backend orchestrates Gemini API calls and patient data management.

Real-world Problem

Family caregivers struggle to:

- Remember multiple medications
- Interpret doctor's prescriptions
- Read discharge summaries
- Coordinate daily care schedules
- Track medication adherence

These challenges often result in medication errors, missed doses, and avoidable hospital readmissions. CareCompass AI simplifies these tasks using AI automation.

Practical Impact

- Reduced medication errors
- Better adherence
- Faster understanding of medical documents
- Lower caregiver stress
- Improved quality of care for elderly patients

Workflow

1. Caregiver uploads prescription image or discharge document.
2. Gemini extracts medicines and care instructions.
3. Backend validates and stores information.
4. AI generates an optimized medication schedule.
5. Caregiver receives reminders and AI assistance.
6. Medication adherence analytics are continuously updated.

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

USP

Unlike conventional reminder applications, CareCompass AI combines **multimodal AI**, **document intelligence**, and **personalized caregiving assistance** into a single platform.

Key differentiators:

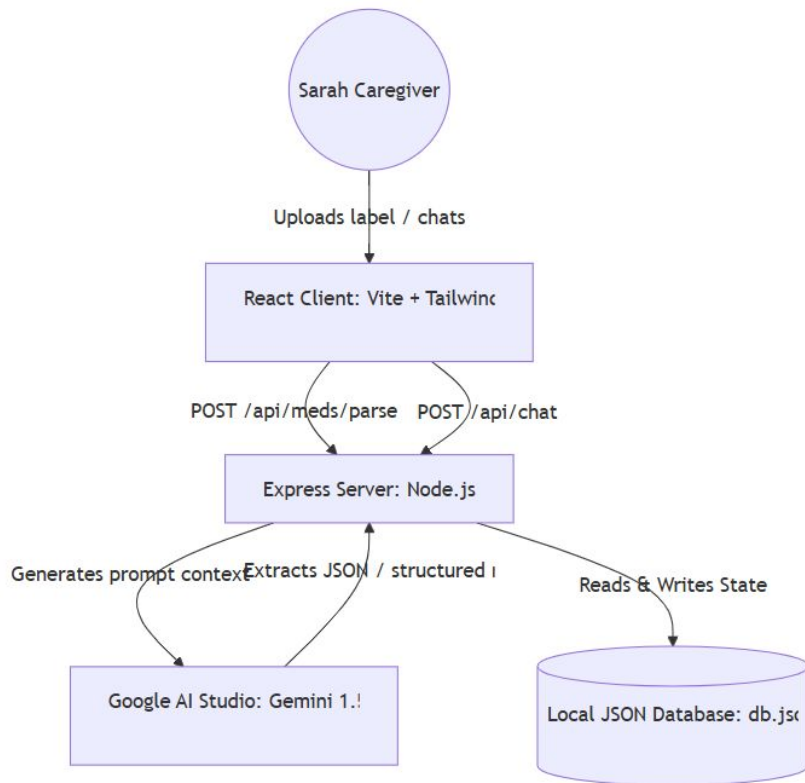
- Prescription OCR using Gemini Vision
- Hospital discharge parsing
- Context-aware AI caregiver chatbot
- AI-generated medication schedules
- Medication adherence analytics
- Integrated eldercare resource search

List of features offered by the solution

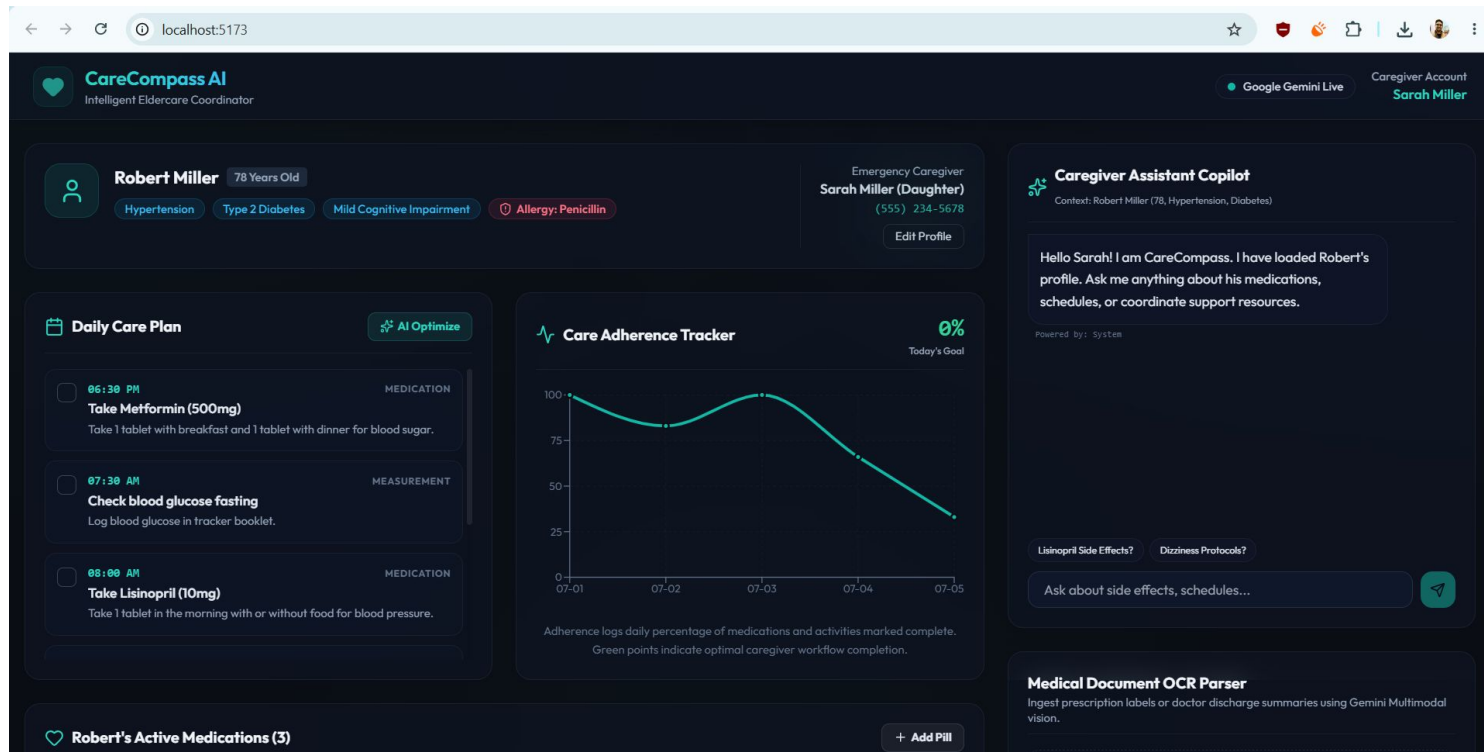
Features

- AI Caregiver Copilot
- Prescription Label OCR
- Hospital Discharge Parser
- Medication Manager
- AI Schedule Optimizer
- Medication Adherence Dashboard
- Eldercare Resource Finder
- Interactive Care Dashboard
- Patient Profile Management
- Real-time AI Assistance

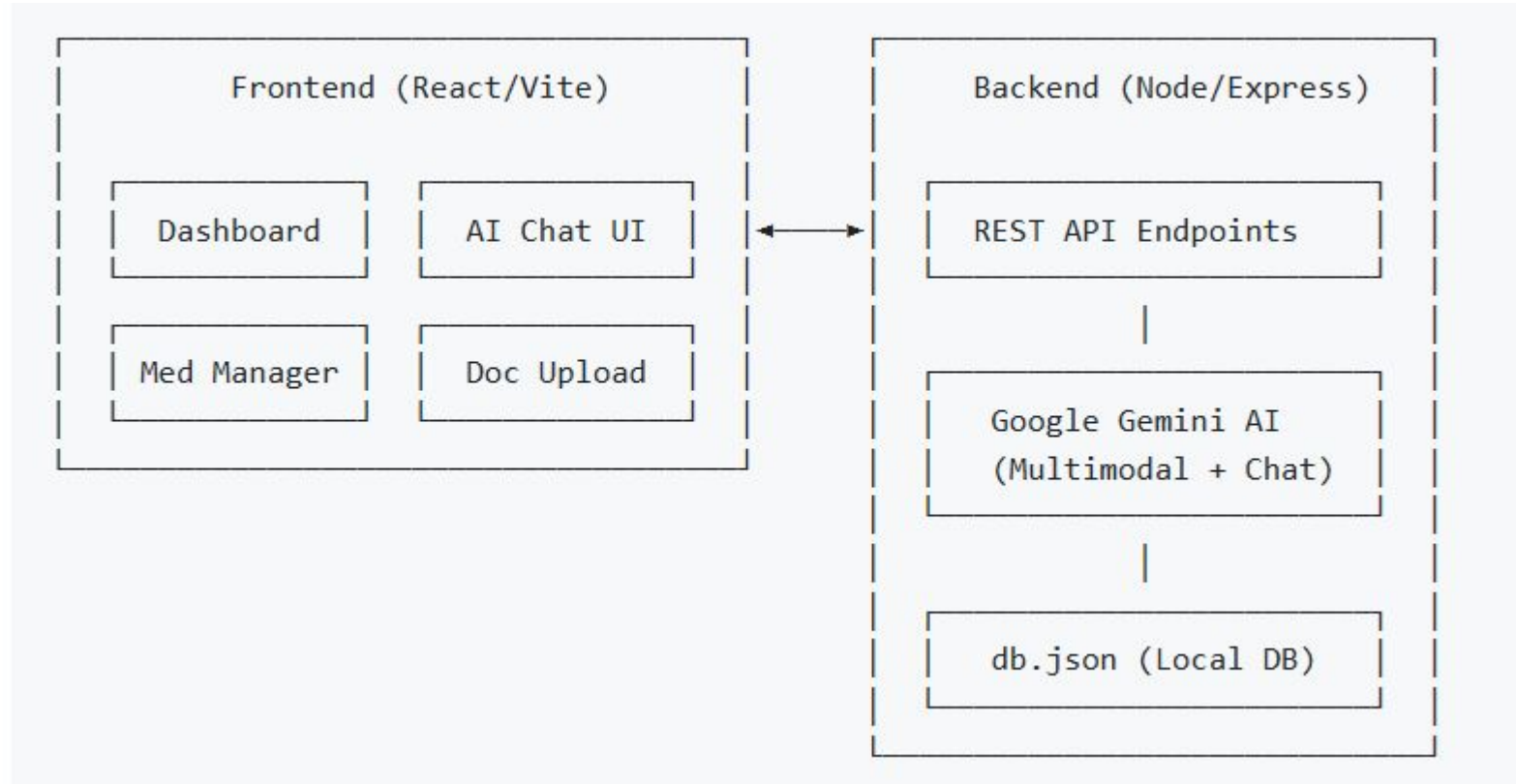
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:



Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Frontend

- React
- Vite
- Tailwind CSS
- Recharts
- Lucide Icons

Database

- JSON Database (Prototype) (Later will use Google Cloud DB / BigQuery)

Deployment

- Google Cloud Run

Backend

- Node.js
- Express.js
- REST APIs

AI

- Google Gemini 3.5 Flash
- Gemini Vision

Snapshots of the prototype



Robert Miller 78 Years Old

Hypertension

Type 2 Diabetes

Mild Cognitive Impairment

Allergy: Penicillin

Emergency Caregiver

Sarah Miller (Daughter)

(555) 234-5678

Edit Profile

 **Daily Care Plan** 

☐ 08:00 AM

Take Lisinopril (10mg)

Blood pressure pill. Take with water.

MEDICATION

☐ 08:30 AM

Take Metformin (500mg)

Blood sugar pill. Take with breakfast.

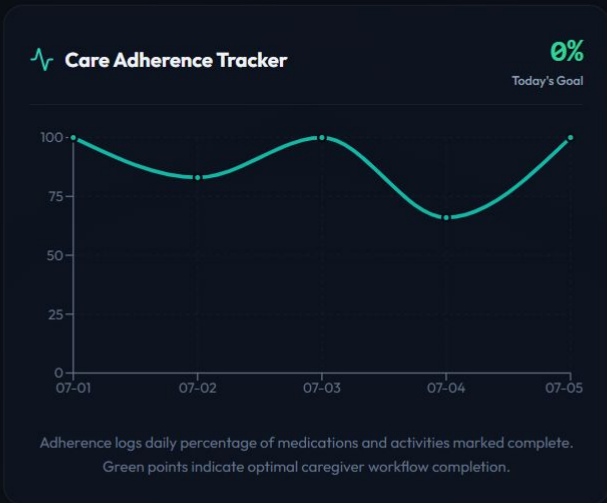
MEDICATION

☐ 09:00 AM

Check Blood Glucose

Target: under 130 mg/dL fasting.

MEASUREMENT



 **Caregiver Assistant Copilot**

Context: Robert Miller (78, Hypertension, Diabetes)

Present with the following issues: medication schedules, or coordinate support resources.

Powered by: System

Are there side effects for Lisinopril?

Robert takes Lisinopril 10mg once daily in the morning (usually at 08:00 AM) for his Hypertension. Make sure he takes it with a glass of water. A known side effect is dizziness or a dry cough. If he experiences dizziness, check his blood pressure immediately.

Powered by: Mock Copilot Engine

Lisinopril Side Effects?

Dizziness Protocols?

Ask about side effects, schedules... 

Medical Document OCR Parser

Ingest prescription labels or doctor discharge summaries using Gemini Multimodal vision.

+ Add Pill

10mg

Once daily

Take 1 tablet in the morning with or without food for blood pressure.

- **Metformin**

500mg

Twice daily

Take 1 tablet with breakfast and 1 tablet with dinner for blood sugar.

- **Atorvastatin**

20mg

Once daily at bedtime

Take 1 tablet at night before bed for cholesterol.

Refills: 2

Ingest prescription labels or doctor discharge summaries using Gemini Multimodal vision.



Upload Med Label Photo



Upload Discharge Slip

Local Eldercare Resource Finder

Search for local clinics, physical therapy, support groups, or senior transportation.

Search resources e.g. wheelchair transit...

Golden Years Eldercare Solutions

© Northside & Downtown

In-home Support

Dedicated caregivers providing non-medical in-home companionship, respite care, light cooking, and hygiene assistance.

☎ (555) 432-1090

Ask AI about them

Silver Wheels Senior Transport

Transportation

Door-to-door medical shuttle service equipped with wheelchair lifts. Offers subsidized



Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Thank you

